

Health Data Sources for Class Projects

ISE 503 Health Analytics, Spring 2026

Data Source Databases

All sources listed are free and accessible within approximately 2 weeks or less. Sources requiring fees or lengthy application processes have been excluded.

Source	URL	Type	Data Level	N (approx.)	Access	Health Areas	Best For
NHANES	https://wwwn.cdc.gov/nchs/nhanes/	U.S. Gov Survey	Individual	~10K/cycle (2-yr cycles)	Open download	Chronic disease, nutrition, biomarkers, demographics	Prediction, risk factor analysis
NHANES - Epidemiologic Follow-up Study (NHEFS)	https://wwwn.cdc.gov/Nchs/Nhanes/Nhefs/Default.aspx Also see: Hernan & Robins, Causal Inference textbook, https://www.hsph.harvard.edu/miguel-hernan/causal-inference-book/	U.S. Gov Survey	Individual	Varies by year, subset of NHANES	Open download	Lifestyle health behaviors and mortality	Causal inference, longitudinal analysis
BRFSS	https://www.cdc.gov/brfss/	U.S. Gov Survey	Individual (state-level sampling)	~400K/yr	Open download	Health behaviors, chronic conditions, preventive care	Classification, state-level comparisons
MEPS	https://meps.ahrq.gov/mepsweb/	U.S. Gov Survey	Individual + household	~35K/yr	Open download	Healthcare utilization, expenditures, insurance	Cost prediction, utilization modeling
CDC WONDER	https://wonder.cdc.gov/	U.S. Gov Registry	Aggregated (county, state, national)	Varies by dataset	Open query tool	Mortality, natality, cancer, environmental exposures	Ecological/population-level analyses

CMS Public Use Files	https://data.cms.gov/	U.S. Gov Claims	Aggregated (provider, geography) + some individual	Millions of records	Open download	Medicare/Medic aid claims, provider quality, spending	Healthcare economics, utilization patterns
SEER	https://seer.cancer.gov/data/	U.S. Gov Registry	Individual	~11M cancer records	Signed DUA (~1 week)	Cancer incidence, survival, treatment, staging	Cancer prediction, survival analysis
NSDUH	https://www.samhsa.gov/data/	U.S. Gov Survey	Individual	~70K/yr	Open download	Substance use, mental health, treatment	Behavioral health prediction
NVSS / Vital Statistics	https://www.cdc.gov/nchs/nvss/index.htm	U.S. Gov Registry	Individual (public-use files)	Millions/yr	Open download	Birth, death, fetal death records	Mortality prediction, maternal/infant health
WHO Global Health Observatory	https://www.who.int/data/gho	International	Aggregated (country)	194 countries	Open download	Life expectancy, disease burden, risk factors, health systems	International comparisons, population health
Global Burden of Disease (IHME)	https://www.healthdata.org/research-analysis/gbd	International	Aggregated (country, region)	195 countries, 350+ diseases	Open download	Death/disability from diseases and risk factors	Global health burden estimation
DHS Program	https://dhsprogram.com/data/	International	Individual + household	Thousands per country, 90+ countries	Free registration (~1 week)	Fertility, nutrition, HIV, maternal health in LMIC	LMIC health analytics, survey analysis
MIMIC-IV	https://physionet.org/content/mimiciv	EHR / Clinical	Individual (ICU patients)	~40K patients	PhysioNet credentialing (~1-2 weeks, CITI training)	ICU care: vitals, labs, meds, diagnoses, notes	Clinical prediction, mortality/readmission models

eICU	https://physionet.org/content/eicu-crd/	EHR / Clinical	Individual (ICU patients)	~200K ICU stays	PhysioNet credentialing (~1-2 weeks)	Multi-center ICU data across U.S. hospitals	Generalizable critical care prediction
PhysioNet (general)	https://physionet.org/	EHR / Clinical / Physiologic	Individual	Varies (dozens of datasets)	Most: free with DUA	Cardiac signals, sleep, clinical databases	Signal processing, clinical prediction
All of Us	https://allofus.nih.gov/	NIH Research Program	Individual	800K+ participants	Free registration + Researcher Workbench onboarding (~1 week)	EHR, surveys, genomics, wearables; emphasizes diversity	Precision medicine, health disparities
UCI ML Repository (Health)	https://archive.ics.uci.edu/datasets?area=Health+and+Medicine	Educational / Published	Individual	Varies (100s-10Ks)	Open download	Heart disease, breast cancer, diabetes, Parkinsons, etc.	Benchmarking, educational projects
Kaggle Health Datasets	https://www.kaggle.com/datasets?tags=13206-Health	Community / Educational	Varies	Varies	Open download	Wide range; verify provenance	Exploratory projects (use caution with unverified sources)
Harvard Dataverse	https://dataverse.harvard.edu/	Research Repository	Individual (study-specific)	Varies	Open download	Published study data across many health domains	Replication, secondary analysis
BioLINCC Teaching Datasets	https://biolincc.nhlbi.nih.gov/teaching	NHLBI / Educational	Individual	~4K (Framingham), ~700 (CAMP), ~7K (DIG)	Free upon request (~days)	Cardiovascular (Framingham), asthma (CAMP), heart failure (DIG)	Teaching, longitudinal analysis

Data Level Key: Individual = subject/patient-level records (one row per person or visit). Aggregated = summarized at a geographic level (county, state, country): not suitable for individual-level prediction but rather for ecological studies and population-level analysis